SOUND & SONG DESIGN

Sound Design: System, Plot and Cue

A ZikShare teaching unit, written for this shelf

Putting sound in a room: the signal path, microphones, playback, the cue plot and the mix that lets an audience hear everything without hearing the system.

WHAT THIS UNIT COVERS

- signal path
- microphones
- playback
- the cue plot
- the mix

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The signal path

Every sound system is the same four things in order: a source (a microphone, a laptop, a phone), a mixer, an amplifier, and a loudspeaker. Learn to trace that path with your finger. When there is no sound, the fault is somewhere on the path, and a designer who can trace it finds the fault in a minute instead of a panic.

Keep the gain structure sane: the source should arrive at the mixer strong but not clipping, and the amplifier should not be asked to make up for a weak signal. Distortion is almost always a gain problem, not a volume problem.

Microphones and the room

A dynamic microphone is robust and forgiving; a condenser is detailed and picks up everything, including the air conditioner. For theatre, body microphones are usually omni-directional lavaliers taped to the hairline or the cheek, with the pack hidden in the costume. Tape the cable in a loop with slack so the actor can turn their head.

Loudspeakers should be in front of the performers, aimed at the audience, never behind them. Feedback is the system hearing itself: move the microphone, lower the gain, or aim the speaker away. The cheapest improvement to most halls is a pair of speakers on stands at the front instead of one box on the floor.

Playback and the cue plot

Use a computer for playback and a backup device for the backup. Name every file with its cue number, convert everything to the same format and sample rate, and test on the actual system - a file that plays at home may not play through the venue's interface.

The cue plot lists number, trigger, file, level, fade in, fade out and destination (which speakers). Write the trigger as something a human can hear: "on her line 'I am going'", not "at 10:42". If the operator cannot follow the script and the plot together, the plot is wrong.

The mix

The audience should hear the voice first, always. Music and effects sit under the voice; if a cue is beautiful and the line is lost, the cue is wrong. Ride the levels during the dress rehearsal rather than setting them once.

Listen from three places: front row centre, the back of the hall, and the wings. The wings are where the actors hear themselves, and if they cannot, they will push and the whole show will get louder.

The show file and the backup

The show file is the sound design in a form the operator can run: every cue numbered, every file named to match, every level set, every fade timed. Save it twice, on the playback machine and on a USB stick, and print the cue list.

Test the whole file on the venue system before the technical rehearsal: every cue, in order, at the level it will play. Files that play at home fail on a venue interface, and the failure is usually a format, a sample rate or a missing codec.

Have a fallback for every cue that matters: a phone with the files, a second cable, a spare battery for the microphone. The audience will forgive a cue that is late; they will not forgive a scene with no sound at all.

Sound effects that convince

A recorded effect convinces when it is layered and when it matches the space. Rain is a bed, a few close drops and a distant muffle; a market is voices at three distances. A single sound file played at one level reads as a sound effect, not as a world.

Fade effects in and out rather than starting and stopping them, and keep a library organised by category with clear names. The cue you need at 11 p.m. is the one you can find in ten seconds.

Exercises

1. Draw the signal path of your venue from microphone to loudspeaker and label every cable.
2. Build a ten-cue plot for a five-minute scene, with triggers and levels.
3. Ring out a room: find the feedback frequency, notch it, and document the settings.

Where to read more

Deena Kaye and James LeBrecht, Sound and Music for the Theatre (2009)

Tomlinson Holman, Sound for Film and Television (2010)

Ric Viers, The Sound Effects Bible (2008)

These are books to look for in a library or a bookshop. They are not on this shelf and ZikShare does not host them.